

1.02 Algebra and Functions

1.02a	Indices	a) Understand and be able to use the laws of indices for all rational exponents. <i>Includes negative and zero indices.</i> <i>Problems may involve the application of more than one of the following laws:</i> $x^a \times x^b = x^{a+b}, x^a \div x^b = x^{a-b}, (x^a)^b = x^{ab}$ $x^{-a} = \frac{1}{x^a}, x^{\frac{m}{n}} = \sqrt[n]{x^m}, x^0 = 1.$		MB1
1.02b	Surds	b) Be able to use and manipulate surds, including rationalising the denominator. <i>Learners should understand and use the equivalence of surd and index notation.</i>		MB2
1.02c	Simultaneous equations	c) Be able to solve simultaneous equations in two variables by elimination and by substitution, including one linear and one quadratic equation. <i>The equations may contain brackets and/or fractions.</i> <i>e.g.</i> $y = 4 - 3x \text{ and } y = x^2 + 2x - 2$ $2xy + y^2 = 4 \text{ and } 2x + 3y = 9$		MB4

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1.02d	Quadratic functions	d) Be able to work with quadratic functions and their graphs, and the discriminant (D or Δ) of a quadratic function, including the conditions for real and repeated roots. <i>i.e. Use the conditions:</i> $b^2 - 4ac > 0 \Rightarrow$ real distinct roots $b^2 - 4ac = 0 \Rightarrow$ repeated roots $b^2 - 4ac < 0 \Rightarrow$ roots are not real <i>to determine the number and nature of the roots of a quadratic equation and relate the results to a graph of the quadratic function.</i>		MB3
1.02e		e) Be able to complete the square of the quadratic polynomial $ax^2 + bx + c$. <i>e.g. Writing $y = ax^2 + bx + c$ in the form $y = a(x + p)^2 + q$ in order to find the line of symmetry $x = -p$, the turning point $(-p, q)$ and to determine the nature of the roots of the equation $ax^2 + bx + c = 0$ for example $2(x + 3)^2 + 4 = 0$ has no real roots because $4 > 0$.</i>		
1.02f		f) Be able to solve quadratic equations including quadratic equations in a function of the unknown. <i>e.g. $x^4 - 5x^2 + 6 = 0$, $x^{\frac{2}{3}} - 5x^{\frac{1}{3}} + 4 = 0$ or</i> $\frac{5}{(2x-1)^2} - \frac{10}{2x-1} = 1.$		

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1.02g	Inequalities	g) Be able to solve linear and quadratic inequalities in a single variable and interpret such inequalities graphically, including inequalities with brackets and fractions. <i>e.g.</i> $10 < 3x + 1 < 16$, $(2x + 5)(x + 3) > 0$. <i>[Quadratic equations with complex roots are excluded.]</i>		MB5
1.02h		h) Be able to express solutions through correct use of 'and' and 'or', or through set notation. <i>Familiarity is expected with the correct use of set notation for intervals, e.g.</i> $\{x : x > 3\}$, $\{x : -2 \leq x \leq 4\}$, $\{x : x > 3\} \cup \{x : -2 \leq x \leq 4\}$, $\{x : x > 3\} \cap \{x : -2 \leq x \leq 4\}$, $\emptyset$. <i>Familiarity is expected with interval notation, e.g.</i> $(2, 3)$, $[2, 3)$ and $[2, \infty)$.		
1.02i		i) Be able to represent linear and quadratic inequalities such as $y > x + 1$ and $y > ax^2 + bx + c$ graphically.		

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1.02j 1.02k	Polynomials	j) Be able to manipulate polynomials algebraically. <i>Includes expanding brackets, collecting like terms, factorising, simple algebraic division and use of the factor theorem.</i> <i>Learners should be familiar with the terms “quadratic”, “cubic” and “parabola”.</i> <i>Learners should be familiar with the factor theorem as:</i> $f(a) = 0 \Leftrightarrow (x - a)$ is a factor of $f(x)$; $f(\frac{b}{a}) = 0 \Leftrightarrow (ax - b)$ is a factor of $f(x)$. <i>They should be able to use the factor theorem to find a linear factor of a polynomial normally of degree ≤ 3. They may also be required to find factors of a polynomial, using any valid method, e.g. by inspection.</i>	k) Be able to simplify rational expressions. <i>Includes factorising and cancelling, and algebraic division by linear expressions.</i> <i>e.g. Rational expressions may be of the form</i> $\frac{x^3 - x - 2}{2x + 1} \text{ or } \frac{(x^2 - x - 6)(x^2 + 4x + 3)}{(x^2 - 9)(x + 3)}.$ <i>Learners should be able to divide a polynomial of degree ≥ 2 by a linear polynomial of the form $(ax - b)$, identify the quotient and remainder and solve equations of degree ≤ 4.</i> <i>The use of the factor theorem and algebraic division may be required.</i>	MB6
1.02l	The modulus function		l) Understand and be able to use the modulus function, including the notation x, and use relations such as $a = b \Leftrightarrow a^2 = b^2$ and $x - a < b \Leftrightarrow a - b < x < a + b$ in the course of solving equations and inequalities. <i>e.g. Solve $x + 2 \leq 2x - 1$.</i>	MB7

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1.02m 1.02s 1.02n 1.02t 1.02o 1.02p 1.02q 1.02r	Curve sketching	m) Understand and be able to use graphs of functions. <i>The difference between plotting and sketching a curve should be known. See Section 2b.</i> n) Be able to sketch curves defined by simple equations including polynomials. <i>e.g. Familiarity is expected with sketching a polynomial of degree ≤ 4 in factorised form, including repeated roots.</i> <i>Sketches may require the determination of stationary points and, where applicable, distinguishing between them.</i> o) Be able to sketch curves defined by $y = \frac{a}{x}$ and $y = \frac{a}{x^2}$ (including their vertical and horizontal asymptotes). p) Be able to interpret the algebraic solution of equations graphically. q) Be able to use intersection points of graphs to solve equations. <i>Intersection points may be between two curves one or more of which may be a polynomial, a trigonometric, an exponential or a reciprocal graph.</i> r) Understand and be able to use proportional relationships and their graphs. <i>i.e. Understand and use different proportional relationships and relate them to linear, reciprocal or other graphs of variation.</i>	s) Be able to sketch the graph of the modulus of a linear function involving a single modulus sign. <i>i.e. Given the graph of $y = ax + b$ sketch the graph of $y = ax + b$.</i> <i>[Graphs of the modulus of other functions are excluded.]</i> t) Be able to solve graphically simple equations and inequalities involving the modulus function.	MB7

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1.02y	Partial fractions		y) Be able to decompose rational functions into partial fractions (denominators not more complicated than squared linear terms and with no more than 3 terms, numerators constant or linear). <i>i.e. The denominator is no more complicated than $(ax + b)(cx + d)^2$ or $(ax + b)(cx + d)(ex + f)$ and the numerator is either a constant or linear term.</i> <i>Learners should be able to use partial fractions with the binomial expansion to find the power series for an algebraic fraction or as part of solving an integration problem.</i>	MB10
1.02z	Models in context		z) Be able to use functions in modelling. <i>Includes consideration of modelling assumptions, limitations and refinements of models, and comparing models.</i>	MB11